

An Unreasonably Cheap Algorithm to Generate Extreme Weather Events

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Motivation

Critical systems and extreme weather

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How to
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- Energy, transport, agriculture: weather dependent
 - Short-to-medium-term operation ← forecasting
 - Long-term planning ← scenario analysis (including extremes)
- Extremes: unlikely but highly relevant for decision-making (potentially disruptive)
- Analyze their consequences → map bottlenecks & points of failure
- Extreme scenarios must be evaluated before any forecast can see them
 - some impacts are unavoidable → response decisions still have to be prepared ahead of the event
 - black-swan type: arguably very highly relevant
- Deciding whether, and how, to prepare for extreme weather involves trading off the cost of preparing for an event which might not occur, against the loss incurred if no preparations are made and it does occur

→ **How can these scenarios be generated?**

Our objective

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Objective: simulate *extreme*, yet *plausible* weather scenarios (in contrast to forecasting)

- Scenarios: simulate weather at different intensity levels
- Example: if temperature goes up by 2° , $\dots$, 6° in this region of interest $\rightarrow$ then $\dots$

Definitions:

- *Extreme*
 - unlikely (and disruptive)
 - lives on the tails of the climatological and forecasting distribution
- *Plausible*
 - transitions are physically grounded
 - variables co-vary in a coherent way

We give a precise description of the scenarios.

Example: stress-testing power-grid control

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- Climate changing fast → summers increasingly disruptive
- Goal: stress-test grid resilience ahead of the coming years
- Focus scenario: **heatwave**
 - co-occurring extremes: heat, solar radiation, wind, fires, thunderstorms, hail
- Potential consequences:
 - grid instability
 - infrastructure damage
 - cascading failures
 - blackouts

How to generate extreme weather events

Our approach

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- **Model:**
 - medium-range generative probabilistic weather model
 - generative: score / diffusion / flow matching
- **Guidance methods:**
 - borrowed from the image generation domain
 - translated to the weather domain
 - two families: sampling-based or gradient-based

But ...

Why not just use the tails of the base forecast distribution?

- Extremes = rare tail events, possibly unprecedented
- May be impossible to obtain from the base forecasting model
- Or simply too expensive to sample
- We want a *distribution* of extremes → even harder
- → need some scheme on top of the base model

Why use a generative model?

- The hope is to keep the generated extremes close to the learned data distribution
- If the modes of the learned data distribution encode physical realism, the generative model can help us achieve extremes that satisfy this property

Images vs. extreme weather

Core challenge of our work: translate / design guidance to match the challenges of the weather domain.

3 key differences:

- **Steps:**
 - image domain → final image obtained one-shot
 - extreme-weather domain → multi-step trajectory (video-alike, but not really ...)
- **Evaluation:**
 - image domain → visual appeal
 - extreme-weather domain → visual checks + quantified realism
- **Target:**
 - image domain → inherent to a specific task (super-resolution, inpainting, prompt alignment, ...)
 - extreme-weather domain → unclear a priori (designed)

Video-alike but not really → transitions must be physically grounded.

Related work

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- guide first step
- shift sampled noise
- plain classifier guidance

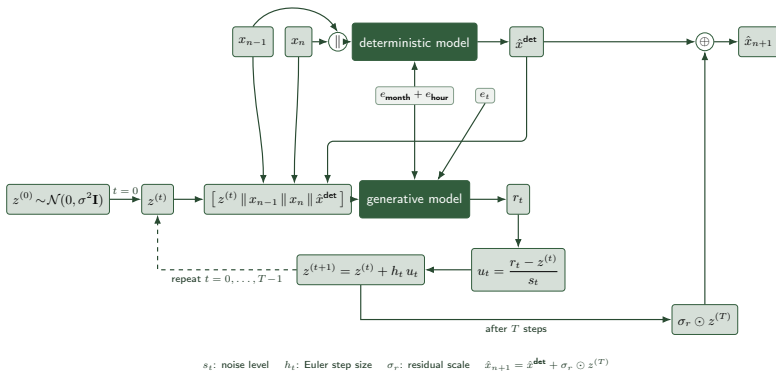
Weather guidance

ERA5 Reanalysis (ECMWF)

- Global, 1.5° grid — 240×121 points
- 82 fields per state:
 - 4 surface: 2m temperature, MSL pressure, 10m u/v wind
 - 6 upper-air $\times$ 13 pressure levels: geopotential, temperature, humidity, u/v wind, vertical velocity
- 24-hour forecast steps, available at 00:00, 06:00, 12:00, 18:00 UTC
- Test set: $\geq 01.01.2020$

Model

Prediction = deterministic model + residual flow matching model



Guidance @ weather time: pipeline

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We target the following:

- heatwave scenarios (surface temperature)
- over a specific region
- with a specified change trajectory
- over a specific timeframe
- with respect to the unguided step

Guidance @ weather time: mask

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Mask:

- Gaussian over the region of interest
- at the variable of interest (surface temperature)

Guidance @ weather time: change trajectory

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Change trajectory:

- over an N -step timeframe
- defined as a percentage change trajectory
- with respect to the unguided step average over the mask
- (later: with respect to the unguided-guided step)

Guidance @ flow time: pipeline

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At every flow step t , we steer the velocity before taking the Euler step:

- 1 Estimate the clean state from the current noisy residual:
$$\hat{x}_t = \hat{x}^{\text{det}} + \sigma_r \odot (z^{(t)} + s_t u_t)$$
- 2 Score it with the masked loss: $L(\hat{x}_t)$
- 3 Take the gradient w.r.t. the noisy state: $g_t = \nabla_{z^{(t)}} L(\hat{x}_t)$
- 4 Steer the velocity: $\tilde{u}_t = u_t - \lambda_t g_t$ with strength $\lambda_t = w \cdot a_t$
- 5 Euler step as usual: $z^{(t+1)} = z^{(t)} + h_t \tilde{u}_t$

Two design choices define the method:

- **Masked loss** — what to push and where
- **Guidance strength-schedule** — how hard and when

Guidance @ flow time: probabilistic view (LBG)

Steering with a loss gradient = sampling from a posterior:

$$p(x|y) \propto p(y|x) p(x)$$

$$\nabla_x \log p(x|y) = \nabla_x \log p(x) + \nabla_x \log p(y|x)$$

Guided vector field

$$\tilde{u}_t = u_t - \lambda_t g_t, \quad g_t = \nabla_{z^{(t)}} L(\hat{x}_t) = -\nabla_{z^{(t)}} \log p(y|z^{(t)})$$

- u_t transports the prior $p(x)$ — the learned weather distribution
- the masked loss defines the likelihood: $p(y|x) \propto e^{-L}$
- exact posterior needs $w = 1$ and a path-specific a_t (e.g. $\frac{s_t}{1-s_t}$ for linear paths)
- we trade exactness for *control*: $\lambda_t = w \cdot a_t$ with the gap-closing schedule
- unlike DPS, no rescaling of $w a_t$ by the gradient norm

Guidance @ flow time: loss

$$L = \left(\sum m \odot \hat{x}_t - (1 + \delta) \sum m \odot x^{\text{ref}} \right)^2$$

- $\sum m \odot x$ — the masked statistic: one number summarizing the region of interest
- $\hat{x}_t$ — the clean-state estimate at flow step t (where we currently are)
- $(1 + \delta) \sum m \odot x^{\text{ref}}$ — the target: the reference statistic, shifted by δ
- x^{ref} — the model's own unguided forecast (or ground truth)
- the signed difference inside the square is the **gap** r_t — the quantity the schedule drives to zero

The gradient flows *through the model* into $z^{(t)}$: the network Jacobian spreads the correction over the whole field.

Guidance @ flow time: strength–schedule

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Applied strength factorizes: $\lambda_t = w \cdot a_t$

- w — global gain: *how hard* to push
- a_t — schedule: *when* to push

Gap closing: each step closes a fraction η of the remaining gap:

$$r_{t+1} = (1 - \eta) r_t$$

This prescribes the residual path $r_t = (1 - \eta)^{t+1} r_0$ and the schedule $a_t = (1 - \eta)^{t+1}$.

Early steps do the heavy lifting; late steps only refine.

Change trajectory

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- We define the full change trajectory at all steps n
- The perturbation paper does it only at $n = 1$, applies the forecasting model N times, and lets the gradient flow back through all N weather steps
- This enables us to control the full trajectory instead, and to have a distribution *over the defined trajectory* — compared to a distribution over possible ways to reach the final target state at N

Guidance methods

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- **OPTIMIZE-STRENGTH**
- **CLOSE-GAP**

OPTIMIZE-STRENGTH

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Optimize the single scalar w against the final loss — no learning rate.

- Run the guided flow with constant w : $\lambda_t = w \cdot a_t$
- w is a scalar, so $\frac{dL}{dw}$ is an *exact* derivative (one adjoint pass)
- Solve $\frac{dL}{dw} = 0$ with the secant method — no learning rate, no iteration count; it stops when the final loss is closed, then reruns once with w^*

Cost: a few full flow evaluations.

CLOSE-GAP

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No w at all — close the gap *exactly* at every step.

- Measure the current gap r_t , aim at the prescribed path:
$$r_t^{\text{target}} = (1 - \eta)^{t+1} r_0$$
- The exact landing is a Newton move along the gradient:

$$\lambda_t = \frac{2 r_t (r_t - r_t^{\text{target}})}{h_t \|g_t\|^2}$$

- Anchored to the *measured* gap $\rightarrow$ drift and overshoot are corrected at the next step
- $\eta = 1$: whole gap in one step; smaller η : spread smoothly over the flow

Cost: a single guided flow — no outer optimization.

Notation summary

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- If we can find metrics that enable us to make statements about the plausibility of the weather states, we don't need to make any assumptions about the physical realism of the guidance target and the learned model function
- If this holds, we can also make statements about full trajectories: if a one-step guided forecast is physically plausible, then so is its autoregressive successor — or better, we can measure the degradation of physical realism
- Reason to guide this way: many little steps that only slightly change the trajectory are good

Experimental setup

Validation

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- Works well for heatwaves because the mask is fixed → not appropriate for cyclones and other moving events
- Sampling methods provide a more direct way to respect the transition function learned by the model → no manual mask and manual trajectories → would side-step the need to check the realism of both mask and delta trajectory, and of the produced results

Future directions

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- GenCast: higher resolution — $0.25^\circ \implies 721 \times 1440$ grid
- Likelihood estimation of an event
- Guiding noise
- Guiding steps

Appendix

Residual model

Appendix

Inputs:

- state — the current analysis x_0 (at t_0)
- prev_state — the state at $t_0 - 24$ h
- pred_state = det_pred — the deterministic prediction
- z_t — the noisy residual latent at the current noise level (the thing being denoised)
- time embedding — month + hour + flow timestep t :
seasonality/diurnal conditioning + noise-level conditioning

Anchoring scenarios in time

Appendix

Option 1: take past events

- extremify them
- compare against available ground truth

Option 2: generate from the present into the future

- the past may not contain all scenarios of interest
- stress-test infrastructure in its current state
- ground truth not available